Indicator Spectral Bands and Logistic Models for Detecting Diesel and Gasoline Polluted Soils Based on Close-Range Hyperspectral Image Data

Jihee Seo, Jaehyung Yu, and Lei Wang

***Abstract*— In this article, we derived indicator spectral bands and classification models for detecting diesel or gasoline pollution in soil using a near- and short-wave-infrared (NIR–SWIR) hyperspectral camera under a close range and laboratory conditions. The soil samples were collected from temperate climate soil with spectral characteristics manifested by secondary minerals. The hyperspectral images show that the diesel and gasoline-polluted soil samples have distinctive spectral differences from clean soil. Different from moisture soil, the spectral absorption features of petroleum hydrocarbons (PHCs) are preserved with an increase in gravimetric content. The more PHCs contents, the stronger the depths at the spectral absorption features. In diesel-polluted soils, the absorption features were observed at various content levels. However, we found a detection limit for gasoline content in the soil, because the absorption features by PHCs disappeared at 8 wt%. To derive the indicator bands, the images were classified by the random forest (RF) algorithm with an accuracy and kappa coefficient of 94.3% and 0.92% using three groups of bands corresponding to ferric ions, C–H stretch/bending, and benzene, toluene, ethylbenzene, and xylene (BTEX) C–H absorptions. The detection models derived from a logistic regression achieved an overall accuracy of 91.82%. The field test of the models on unprocessed soils achieved an accuracy of 83.36%. Because of their simple forms, the logistic detection models can be transferred to remote sensing applications of soil PHC pollution under close-range conditions such as drone-based projects.**

***Index Terms*— Hyperspectral spectroscopy, indicator spectral bands, petroleum hydrocarbons (PHCs) pollution, soil survey.**

I. Introduction

SOIL pollution induced by anthropogenic activities has been a serious socio-economic issue around the world [1]. Specifically, soil pollution caused by petroleum products is mainly associated with industrial and causes severe problems because of its neurotoxic effects on humans and animals [2]. Long-term exposure to petroleum hydrocarbons (PHCs hereafter) could increase the risk of lung, skin, and bladder cancers [3], [4]. Therefore, effective identification of PHC pollution in the soil is critical to establish the environmental restoration actions that many countries are seeking.

Traditionally, petroleum-polluted soils are sampled from the field and analyzed in laboratory conditions mainly using gas chromatography (GC) to assess the type and intensity of soil pollution. The GC method requires the preprocessing of samples by extracting the pollutant with chemical solvents, and the pollution level is measured with a flame ionization detector (FID) [5]. Although the GC method is accurate, the drawback is that it is very expensive and time-consuming [6], [7], [8], [9], [10].

As an alternative to GC, infrared spectroscopy provides a time and cost-effective solution [6], [9]. Many spectroscopic analyses have been tested for the detection of total petroleum hydrocarbons (TPHs) concentration in soil. Chakraborty et al. [10] conducted a regression analysis using reflectance spectroscopy between 350 and 2500 nm to predict TPH content from crude oil-polluted soils in an oil field with an R^2 of 0.64. Chakraborty et al. [6] combined reflectance spectroscopy (350–2500 nm) with portable X-ray fluorescence spectrometry (PXRF) for rapid and improved quantification of soil petroleum contamination with a better R^2 of 0.78. Ng et al. [11] employed mid-infrared (MIR) (2500–25 000 nm) and near-infrared (NIR)—shortwave-infrared (SWIR) (780–2500 nm) spectroscopy and developed prediction models for TPH concentration for crude oil and diesel polluted soils based on multiple linear regression (MLR) and linear partial least squares regression (PLSR) models using spectral bands from 3344–3559 to 2300–2340 nm. Douglas et al. [12] compared the performance of PLSR with a nonlinear random forest (RF) regression for the calibration of a VIS–NIR model when analyzing TPH in soils.

With the development of hyperspectral imaging systems, a few imaging approaches for PHCs detection have been introduced. Compared to spectroscopic analysis, hyperspectral imaging systems have the advantage of bulk sample processing for the detection of pollution distribution. Pelta and Ben-Dor [13] tested the feasibility of a close-range SWIR hyperspectral camera (950–2500 nm) for the detection of PHCs in crude oil-contaminated soils and found that the detection limit in the field is not only controlled by the PHCs

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proportion but also by soil types. Correa Pabón et al. [14] used VNIR–SWIR spectroscopic analysis and airborne hyperspectral images to classify crude oil-contaminated soils in crude oil storage facilities using Multiple Endmember Spectral Mixture Analysis (MESMA) with an accuracy of 98%.

The previous studies have reported that the main spectral features of PHCs occur in the 1700- and 2300-nm regions, especially in the near- and shortwave-infrared (NIR–SWIR) spectral region (1000–2500 nm) [15], [16]. The spectral features vary by type of PHCs. Crude oil has an asymmetric absorption feature around 1209 nm and a double absorption feature around 1725 and 1757 nm. Diesel has similar doublet absorption with some minor absorption features found at 2177, 1155, 1202, and 1640 nm. Gasoline shows different spectral features from crude oil and diesel, with major absorption features around 1700–1774 nm and 2280–2450 nm [17], [18], [19].

Previous studies mostly focused on pollution caused by crude oil in oil fields or near large oil storage facilities [11], [12], [13]. Moreover, there was a lack of studies to report the detection limit of oil content in the soil that could vary based on different oil types (e.g., gasoline and diesel) and soil. Those parameters are critical components for real-world applications [8], [11], [20], [21]. Although crude oil pollution is still the dominant type of soil contamination by PHCs, refined oil products are more of the concern to social and public health researchers because gasoline tank leaks or spills have a more direct impact on people's day-to-day lives and health conditions. Furthermore, the comparison between soil moisture and PHCs pollution was often ignored in the previous study of PHCs pollution detection, whereas in real-world cases soil moisture could be confused with PHCs pollution.

The objectives of this article are to: 1) identify the effective indicator spectral bands for gasoline and diesel pollution detection under the influence of mineral composition and water content and 2) establish a simple logistic regression model to classify a close-range hyperspectral image to polluted and nonpolluted segments based on the model detection limit. We used a NIR–SWIR hyperspectral camera for spectral and imaging analysis. Samples were controlled by varying the percent of gasoline, diesel, and water mixed with soil.

II. Materials and Methods

A. Sample Collection and Preparation

We collected 20 kg of soil samples for the gravimetric content analysis of gasoline, diesel, and water from a test field at Chungnam National University, Daejeon, South Korea. To figure out spectral variations associated with the content variation, possible spectral variables including the effects of organic materials, moisture, and granules were removed. The soil samples were collected at a depth of 15 cm or below after removing humus layers resulting minimization of organic components. The samples were transported in polyethylene bags to the laboratory and completely dried to avoid the moisture effects. The dried samples were sieved to select soils under 100 meshes to remove the granule effect. The preprocessed soils were split into 30 g of samples and positioned in petri-dishes. As a result, a total of 114 samples were prepared for gravimetric content experiment and hyperspectral data acquisition. Diesel and gasoline oils were obtained from GS Caltex Corporation, South Korea.

B. Mineralogical Analysis

Soil is composed of minerals, organic materials, moisture, and air. Therefore, its spectral characteristics are manifested by those components. Given that the organic materials and moisture effects were minimized in this study, the spectral characteristics of the preprocessed soils are only controlled by mineral composition. Before the samples were treated in further experiments, an X-ray diffraction (XRD) analysis was conducted to figure out the mineral composition of the samples. A total of five representative samples were selected for the XRD analysis based on the visual inspection, and the results were used for the interpretation of spectral characteristics manifested by specific minerals. The XRD analysis used a Rigaku Ultima IV X-ray diffractometer with Cu–Kα radiation ($\lambda = 1.5406$ Å), where the X-ray tube voltage and current were set to 40 kV and 30 mA, respectively, and the diffraction patterns were acquired with a 2θ range of 3° to –90° at a scan step of 0.02° and a scan speed of 20°/min.

C. Gravimetric Content Experiment

The 30 g of preprocessed soil samples were separated and spread in a 9-cm-diameter glass petri dish with a 4 mm thickness. This study tries to derive models for the detection of gasoline and diesel-polluted soils. We conducted gravimetric experiments on dry soils with various amounts of water, gasoline, and diesel, respectively. Among the 114 soil samples, 31, 41, and 42 samples were used for water content, gasoline content, and diesel content experiments, respectively. The samples were fully saturated with distilled water, gasoline, and diesel, and air-dried. As the samples were gradually dried, the length of drying time and weight of samples were measured to record the drying speed and gravimetric content of water, gasoline, and diesel in soils. The measurements were made at least 5 times per sample depending on the drying speed. At each level of gravimetric content, including the dry condition, hyperspectral images were acquired to assess the spectral variations associated with various gravimetric contents. As a result, a total of 612 measurements were made at various content levels of the three compounds mixed into the 114 soil samples. The gravimetric content $[W]$ of water, gasoline, and diesel in soils was calculated by the following equation:

$$W = \frac{m - m_0}{m_0} \times 100\% \tag{1}$$

where m is the weight of the wet sample (g), and m_0 is the weight of the dry sample (g).

D. Hyperspectral Image Acquisition and Preprocessing

The hyperspectral images were captured using a Specim hyperspectral SWIR camera (Spectral Imaging Ltd., Finland) in laboratory conditions. The camera has a spectral range of

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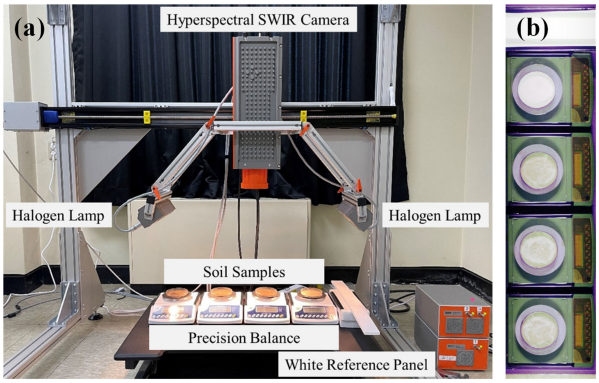


Fig. 1. (a) Hyperspectral image acquisition of soil samples using Specim SWIR camera at art scanning mode and (b) example of the scanned image.

TABLE I
TECHNICAL SPECIFICATION FOR SPECIM SWIR HYPERSPECTRAL CAMERA

Spectral range	916nm-2500nm	
Spectral bands	288	
Spectral resolution	12nm	
Spectral sampling	5.6nm	
Spatial resolution	384 pixels	
Pixel Size	24*24μm	
FOV	23°	
Frame rate	450fps (maximum full frame)	
SNR	1050:1 (at max, signal level)	
Detector	Cryogenically cooled MCT detector	
Power consumption	Nominal< 50W	
Size(l*w*h)	Sensor	Power supply& Control Unit
	470*176*178nm	300*190*130nm
Weight	14kg	5kg

916–2500 nm (NIR–SWIR) with a 12-nm spectral interval and 5.6-nm spectral sampling, producing hyperspectral images of 288 bands for 384 pixels. For the Lambertian reflectance data acquisition, the samples were leveled to the camera at a nadir view, with a halogen lamp as the light source (see Fig. 1) [23].

The petri dishes were scanned with a horizontal line scanner mode with two halogen light sources positioned from an 88 cm distance. A white reference panel (Spectralon material with 99% reflectance) was stationed next to the samples in the field of view (FOV) for radiometric calibration [23], [24]. The image acquisition parameters for the hyperspectral imaging system were 16-ms exposure time, 40-Hz frame rate, one spectral binning, one spatial binning, and 23° FOV (see Table I). The image acquisition system was controlled by the Lumo Scanner v. 2.6 software (Specim Ltd., Finland). Prior to each measurement, dark images were automatically recorded and stored with a closed shutter for additional radiometric calibration. The acquired hyperspectral reflectance images were spectrally smoothed using a filter with a width of two spectral intervals and a second-order polynomial smoothing function using the Tactical Hyperspectral Operations Resource (THOR) Spectral Smoothing tools of the Environment for Visualizing Images (ENVI).

In addition, we applied the maximum noise fraction (MNF) transformation to remove random noise in the hyperspectral data [25]. The noisy bands were determined by the eigenvalues less than or equal to 2. Previous studies (see [26]) suggested that cut-off eigenvalues of 2 for maximum noise removal without disturbance of the original data. The noisy bands were then replaced with zero values, and the data were transformed back to the spectral domain by the inverse MNF transformation. After the noise removal, the background pixels were excluded to analyze the spectral characteristics of soil samples using the masking tool in ENVI. To avoid the impact by water vapor in the field conditions, the spectral bands associated with water absorption features (1343–1605 and 1800–2148 nm) were excluded, and, thus, 155 of the 288 bands were used for further analyses.

E. Spectral Analysis

The preprocessed hyperspectral images of the soil samples with different levels of gasoline, diesel, and water content were used for spectral analysis. The reflectance spectra were transformed into hull-quotient spectra to measure changes in the absorption features of the soil samples. The hull-quotient transform enhances absorption features and helps in the detection of absorption positions and absorption depths [27], [28], [29]. The continuum is removed by dividing the original reflectance values in an absorption trough by the corresponding values of the continuum line [27]. The output curves have values between 0 and 1 [30].

The reflectance and hull-quotient spectra of dried soil samples were analyzed for spectral characteristics associated with the mineral composition and used for control spectra for the detection of spectral variations by the addition of water, gasoline, and diesel. Moreover, the spectral variations associated with various gravimetric contents of water, gasoline, and diesel were analyzed using hyperspectral images of water, gasoline, and diesel soil samples. Then, spectral variations of soil samples associated with different levels of gravimetric content in water, gasoline, and diesel were analyzed. The spectral analyses were carried out in ENVI 5.3 and The Spectral Geologist (TSG) 7.5 software. The spectral features were compared with library spectra to figure out the spectral characteristics associated with mineral composition. The spectral libraries include the United States Geological Survey (USGS) Spectral Library Version 6 and Version 7 and the Advanced Space-borne Thermal Emission and Reflectance Radiometer (ASTER) Spectral Library Version 2.0 of the Jet Propulsion Laboratory (JPL).

F. Calibration and Validation Data Selection

This study classified polluted soils using two different approaches: RF classification algorithm and binary logistic regression model. The RF is one of the machine learning algorithms that give satisfactory classification results among the machine learning algorithms. Although the machine learning algorithm provides better classification accuracy, the RF cannot be transferable for other cases without additional training data. In contrast, the logistic regression models are fully transferable because their equations and coefficients are defined.

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TABLE II
NUMBER OF SAMPLES AND PIXELS DIVIDED INTO CALIBRATION AND VALIDATION GROUP

No. of Samples	Calibration (70%)	Validation (30%)	Total
Dry	107	46	153
Water	93	39	132
Diesel	129	57	186
Gasoline	99	42	141
Total	428	184	612
No. of Pixels ((No. of Samples)× 60)	Calibration (70%)	Validation (30%)	Total
Dry	6420	2760	9180
Water	5580	2340	7920
Diesel	7740	3420	11160
Gasoline	5940	2520	8460
Total	25680	11040	36720

A total of 612 hyperspectral spectral images including 153 dry soil, 132 water soil, 141 gasoline soil, and 186 diesel soil with known gravimetric contents were used to generate training and validation datasets for classification models (see Table II). From each hyperspectral pixels were randomly selected as data to train and validate the classification models. The total number of pixels for each soil class is listed in the "total" column of Table II. For each class, 70% of the pixels were used for training, and 30% of the pixels were used for validation.

As a result, a total of 36 720 soil spectra were extracted from the hyperspectral images and classified into four classes including dry soil, moist soil (water soil hereafter), gasoline-contaminated soil (gasoline soil hereafter), and diesel-contaminated soil (diesel soil hereafter). The gravimetric content of the water, gasoline, and diesel in the soil samples ranged from 0.1% to 72.4%, 1.6% to 53.7%, and 31.8% to 57.3%, respectively.

G. Random Forest (RF) Machine Learning

The machine learning model (RF) [31], [32], [33] was used to learn from the training sample data for assessing the separability of the classes and ranking the spectral bands by their contribution to the classification power (Gini index).

The RF model uses bagging and bootstrapping to develop decision tree ensembles [34]. The bagging and bootstrapping select 2/3 of samples as the training set known as "in-bag" and reserve the rest 1/3 of samples as the test set known as "out-of-bag (OOB)." The classification accuracy is assessed by cross-validation between training and testing sets. The iterative generation of decision trees is formed based on Ntree (number of trees to grow) and m_{try} (number of variables to divide at each note), and the final classification decision is made by the majority votes from the ensemble [31], [35].

RF provides variable importance assessment by the Gini index and mean decrease accuracy [36]. The Gini index, also called the impurity index, is used for testing how important a variable is in a decision tree. The mean decrease accuracy evaluates the decrease in accuracy by each variable, where the larger decrease indicates the higher importance of the variable [37]. The higher the mean decrease in Gini value, the more important the variable in the classification model.

The RF classification results were assessed by confusion matrices and the top 30 bands with the highest Gini indices were used as candidate spectral indicators for water, gasoline, and diesel content detection [38]. The spectral characteristics of those 30 bands in association with PHCs and mineral components were analyzed. The RF classification was carried out in the Rstudio software. Although machine learning models are more robust for image classification, they are difficult to use in different case studies, especially by different groups of researchers. Simple models, such as the logistic regression presented in Section II-H, can be easily transferred and updated, especially for field conditions.

H. Detection Model Development and Evaluation

The binary logistic regression models for the detection of detecting four soil classes in the hyperspectral images were developed (Fig. 2). A binary logistic regression model is a statistical technique used to predict the probability of an event using a linear combination of independent variables [39], [40]. Therefore, the logistic regression model is easy to deploy and generalizable.

1) Band Selection and Detection Model Derivation: The detection models for water, gasoline, and diesel in soil samples were derived by logistic regression models using the NIR–SWIR bands excluding spectral bands with reduced sensor sensibility (971–1027 nm) and water absorption bands [41]. The purpose is to make the logistic models generalizable to field conditions.

The binary logistic regression is composed of one or multiple independent variables associated with a dependent response variable [42], [43]. The equation can be written as a function of P for dry, water, diesel, or gasoline contents, respectively, as follows:

$$P_{\text{dry, water, diesel or gasoline soil}} = \frac{e^{(C+\beta_1 x_1+\beta_2 x_2+\cdots+\beta_n x_n)}}{1+e^{(C+\beta_1 x_1+\beta_2 x_2+\cdots+\beta_n x_n)}} \quad (2)$$

where $P_{\text{dry, water, diesel or gasoline soil}}$ is the probability of a positive occurrence of each class, C is the constant (or intercept) of the equation, β is the coefficient (or beta weight) of the predictor variables, and x is the reflectance value of the selected spectral band.

2) Model Evaluation: The models are evaluated by Hosmer and Lemeshow tests as well as the Wald statistics. In addition, pseudo-R^2 values of "Nagelkerke" and "Cox and Snell" were additionally used for model validation. The Hosmer and Lemeshow tests are goodness-of-fit tests for logistic regression to evaluate the model based on a log-likelihood ratio between the observation and prediction values. The tests divide the data into deciles repeating tenfold and assess the model performance in each subgroup by the likelihood of matching between observed and predicted classes. Then, the test statistic H is computed with a p-value to assess the model's prediction [44].

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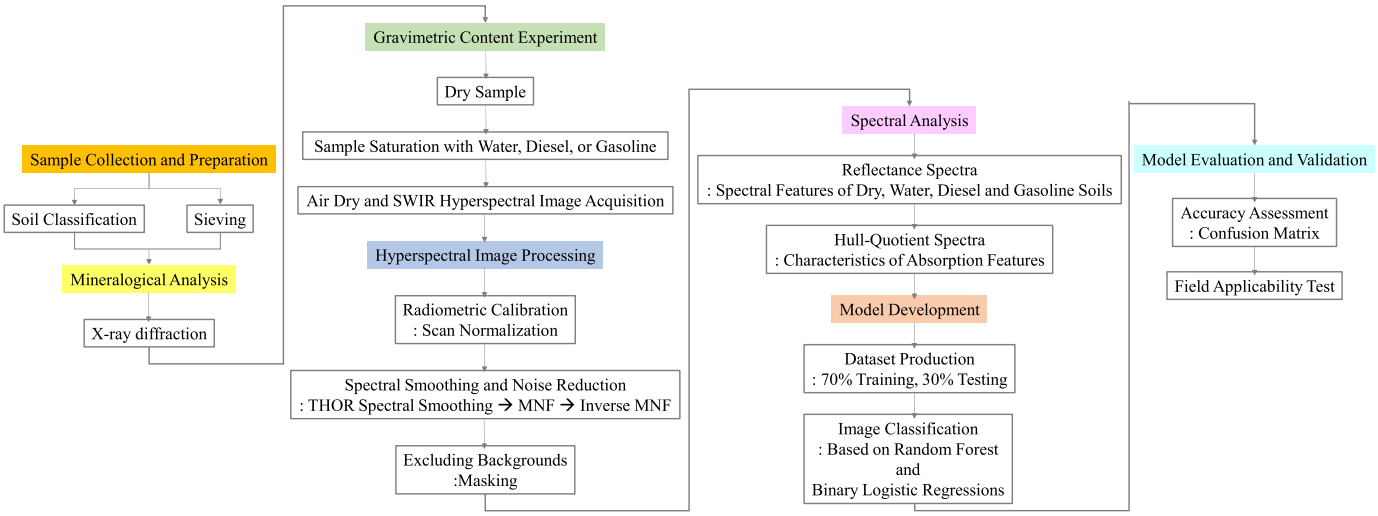


Fig. 2. Schematic of methodology.

Moreover, Cox–Snell and Nagelkerke R^2 statistics were calculated by the following equations [45], [46], [47]:

$$R^2 = 1 - \left\{ \frac{L(M_{\text{Intercept}})}{L(M_{\text{Full}})} \right\}^{\frac{2}{N}}, \quad \text{Cox and Snell} \tag{3}$$

$$R^2 = \frac{1 - \left\{ \frac{L(M_{\text{Intercept}})}{L(M_{\text{Full}})} \right\}^{\frac{2}{N}}}{1 - L\left(M_{\text{Intercept}}\right)^{\frac{2}{N}}}, \quad \text{Nagelkerke} \tag{4}$$

where $L(M_{\text{Intercept}})$ is the log-likelihood for the model without explanatory variables, and $L(M_{\text{Full}})$ is the log-likelihood for the model with explanatory variables. The ratio of the pseudo-R^2 closer to 1 indicates the better model effectiveness [48]. Also, the Wald statistic [(coefficient/standard error)^2] was used to evaluate the statistical significance of each explanatory variable [49]. As a result, the best equation for each target was selected with the highest statistical significance.

The derived logistic regression models for each class are applied to the hyperspectral images using the "bandmath" tool in ENVI software. The image classification results were assessed by confusion matrices, overall accuracy, kappa coefficient, commission error, and omission error.

3) Field Applicability Test: To assess the field applicability of detection models derived from the binary logistic regression models, an outdoor experiment with a natural illumination condition was conducted to simulate a real-world case in the field. Different from the laboratory experiment, the outdoor experiment used raw soils with one-day air dry. The soil from the test field was spread out to 4.4 m^2 and soaked with water, diesel, and gasoline at three different sectors (see Fig. 3). The rest area was left for dry soil.

The Specim hyperspectral SWIR camera was positioned on a tripod at an off-nadir-looking angle to capture the image. The target was scanned four times at an 8-m distance. A white barium sulfate reference panel was stationed in the FOV next to the target soils for radiometric calibration. The exposure time and the camera focus were adjusted to avoid oversaturation. The images were acquired from 13:00 to 15:00 on 25 February 2022. After radiometric calibration, the image

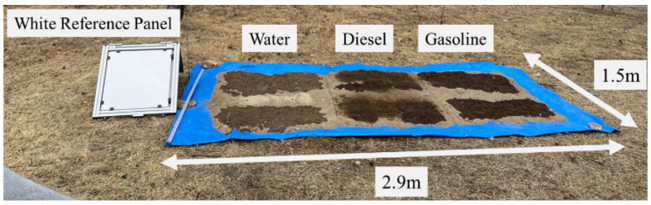


Fig. 3. Outdoor experiment setting to test field applicability.

with the best quality was preprocessed to surface reflectance and fed into the logistic regression models for classification. Finally, an accuracy assessment was conducted.

III. Results and Discussion

A. Mineral Composition and USDA Soil Classification

The spectral properties of soils are mainly controlled by their components including air, moisture, minerals, and organic materials. The most impacting components are minerals because their proportion is over 90% among solid materials, excluding air and water. Based on XRD analysis, the primary soil minerals are quartz, K-feldspar, calcite, and muscovite and secondary minerals are illite, kaolinite, montmorillonite, and hematite. In general, the primary minerals in soils are composed of silicate minerals, which are relatively stronger against weathering processes [50]. Therefore, the commonly found primary minerals in soils are quartz followed by muscovite, K-feldspar, and calcite [51]. The secondary minerals are products of the chemical weathering process where the hydrolysis process of silicate minerals generates Al, Fe, and Mg and OH combined minerals. Given the mineral composition of the soil samples, the soils used in this study can be considered representative soils that originated from the weathering process of temperate climate [52], [53], [54], [55], [56].

The soil analysis found that the samples were composed of sand, silt, and clay by 76%, 15%, and 9%, respectively (see Fig. 4). It concludes that the soils can be classified as sandy loam soil based on the United States Department

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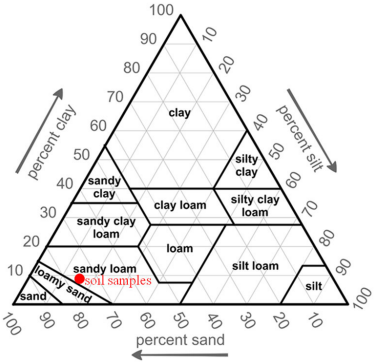


Fig. 4. Definition of soil class based on the USDA texture triangle (USDA, 2017).

of Agriculture (USDA) soil classification system (USDA, 2017) [57]. This soil type is commonly found in the natural environment [58], and, thus, can be considered representative of real-world cases.

B. Spectral Characteristics of Dry, Water, Diesel, and Gasoline Soils

1) Dry Soil: The reflectance curve of dry soil samples shows strong absorption features near 1410–1450, 1915, and 2208 nm and some weak absorption features near 1200 and 2400 nm [see Fig. 5(a)]. The absorption features near 1410–1450 and 1915 nm are associated with chemical components of hydroxyl (OH) and water ions. Considering the mineral composition of the soil samples, the absorption feature at 2208 nm is manifested by clay minerals with chemical components of AlOH such as smectite, illite, kaolinite, or montmorillonite. In addition, the absorption features near 1200 nm are manifested by ferric iron related to iron oxides, and those near 2400 nm are manifested by MgOH or $CaCO_3$ related to calcite [59], [60].

2) Water Soil: The spectral characteristics of soil samples with various water contents were analyzed from fully saturated soil (64% moisture content) to dry soil (0.1% moisture content). In general, the spectral features of soil samples with water showed similar absorptions to the dry samples while the absorption depths are altered by their water content. However, the overall reflectance decreased with an increase in water content, and the absorption depths associated water ions at 1410 and 1910 nm increased significantly. At the same time, the changes in clay absorption by AlOH/MgOH and ferric ion absorption by Fe^{2+}/Fe^{3+} were insignificant with water content. The results confirm that the spectral variations associated with the moisture content and AlOH or ferric irons in soils do not have significant correlations, as suggested by the previous study [20].

3) Diesel Soil: It took a long time for diesel to evaporate from the diesel soil samples, as shown in previous studies [61]. There was only a 20% of reduction in the diesel content during 80 days of air dry in our experiment. Therefore, the spectral characteristics can only be obtained from gravimetric diesel content ranging from 54% to 33%.

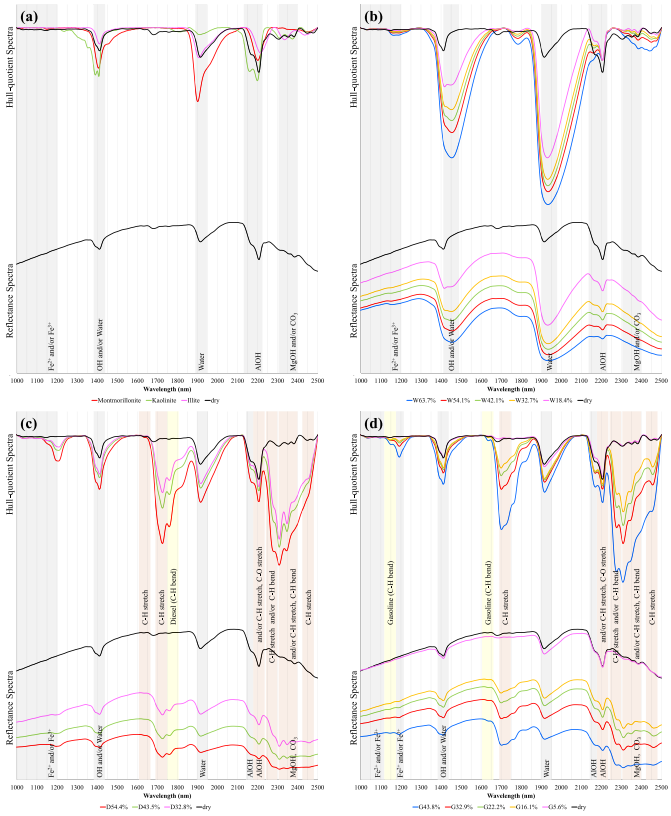


Fig. 5. Representative SWIR reflectance and hull-quotient spectra (offset for clarity) of the dry soil samples and reference spectra of (a) montmorillonite, kaolinite, and illite, (b) soil samples containing various gravimetric content of water, (c) diesel, and (d) gasoline, respectively.

The spectral signatures of diesel soils showed integrative spectral characteristics manifested by both soils and PHCs [see Fig. 5(c)]. The absorption features manifested by secondary minerals and moisture effect were detected at all levels of diesel content [see Fig. 5(c)]. In addition to the mineral absorptions, different from dry or water soils, the diesel soils showed distinctive PHCs absorption features near 1720–1770, 2200, and 2250–2450 nm. These absorption features are attributed to C–H stretching or bending modes of terminal CH_3 and saturated CH_2 groups linked to PHCs [62], [63], [64], [65]. Those spectral features are caused by combinations or overtones of C–H stretching vibrations of saturated CH_2 and CH_3 groups in addition to methylenic, olefinic, or aromatic C–H functional groups [62].

The absorption depths of PHCs absorptions systematically increased with the increase in diesel content. It indicates that the chemical components of C–H associated with PHCs increased with more diesel contents. It is observed that the absorption depths associated with secondary minerals also increased with an increase in diesel content even if the variation was not as significant as PHCs absorptions [see Fig. 5(c)]. Different from water soils, the chemical component of diesel will not interfere with the spectral characteristics of dry soils. Indeed, the chemical composition report of diesel by the refinery company listed PHCs as the major component with a minimal amount of water and residues composed of phosphate, sulfate, oxide, and silicate [66]. The spectral features of the soil samples are associated with oxides and

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hydro-oxides, where the spectral features might overlap with diesel features because of similar chemical components [20].

4) Gasoline Soil: The spectral characteristics of gasoline soils were analyzed for gravimetric content from 5% to 44%. Notably, the evaporation rate of gasoline was faster than other types, where the gravimetric content decreased as low as 8% within 4 h after the saturation. The overall spectral patterns of gasoline soils are similar to those of diesel soils showing overall reflectance decrease and increase in absorption depth at both minerals-associated and PHCs-associated spectral features with an increase in gasoline content [see Fig. 5(d)]. Compared with the spectral characteristics of diesel soils, the gasoline soils showed relatively weaker doublet absorptions near 1750 and 2350 nm, where the spectral ranges are associated with the diesel C–H bending effect. However, the doublet absorption feature at 2270 nm and the absorption feature near 2450 nm were stronger [see Fig. 5(c) and (d)]. In addition, the weak absorption features near 1630 and 1140 nm were detected only in gasoline soils manifested by C–H bending effect of gasoline [see Fig. 5(d)]. The differences can be explained by the gasoline production procedure, where gasoline contains a high concentration of aromatic hydrocarbons associated with aromatic C–H bends at 1140 and 1630 nm [67]. Moreover, gasoline contains more benzene, toluene, ethylbenzene, and xylene (BTEX) than diesel and BTEX creates critical absorption features beyond 2400 nm [61], [68]. Indeed, the previous studies showed that the specific absorption minima of petroleum at ∼1645 nm was derived from C–H stretching modes of ArCH linked to polycyclic aromatic hydrocarbons, and ∼1752 nm was associated with the C–H stretching mode of saturated CH_2 group in the first overtone region. Also, the spectral features near 2240 nm were due to the stretch and/or bend of C–H in the NIR region [64], [69].

Notably, the spectral features associated with PHCs in gasoline soils disappeared when the gravimetric content was 5.6% [see Fig. 5(d)]. Because the PHCs feature is present for the 16.1% spectral curve, it indicates that the detection threshold of gasoline in soils using spectral variables exists between 16.1% and 5.6%. We checked all spectra of gasoline soils and found that the disappearance occurred when the gravimetric content was below 8%. This phenomenon was not reported in previous studies.

C. Classification of Soil Samples and Derivation of Detection Models

1) Random Forest Classification: The RF was carried out with Ntrees = 290 and m_{try} = 78 based on the OOB error analysis (see Fig. 6). The RF classification showed that the spectral signals acquired from the hyperspectral camera can be effectively used for the detection of water, diesel, and gasoline soils showing the overall accuracy and kappa coefficient of 94.3% and 0.92% (see Table III), respectively. The classification accuracy was highest for water, followed by dry, diesel, and gasoline classes.

Although the accuracy was higher than 90% for most cases, the producer's accuracy for gasoline soils was relatively low

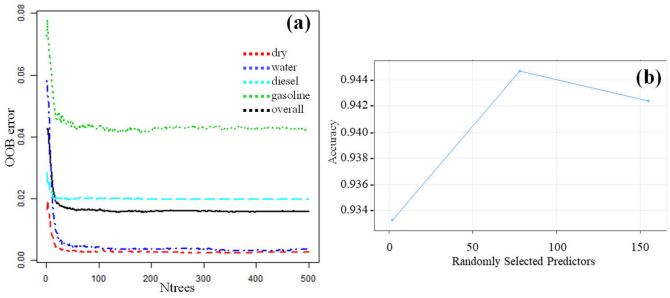


Fig. 6. Variation of (a) OOB error of Ntrees and (b) accuracy of m_{try}.

TABLE III

CONFUSION MATRIX TABLE OF THE TRAINING AND VALIDATION DATA FOR THE CLASSIFICATION RESULTS FROM THE RF CLASSIFICATION

Training Data					
Class	User's Accuracy (%)	Producer's Accuracy (%)	Accuracy (%)	Commission Error (%)	Omission Error (%)
Dry	92.0	98.8	98.8	8.0	1.2
Water	97.6	96.9	96.9	2.4	3.1
Diesel	95.0	95.8	95.8	5.0	4.2
Gasoline	93.1	85.1	85.1	6.9	14.9
Overall Accuracy: 94.3%					
Kappa Coefficient: 0.92					

Validation Data					
Class	User's Accuracy (%)	Producer's Accuracy (%)	Accuracy (%)	Commission Error (%)	Omission Error (%)
Dry	92.6	98.6	98.6	7.4	1.4
Water	97.5	97.9	97.9	2.5	2.1
Diesel	95.2	94.6	94.6	4.8	5.4
Gasoline	91.6	85.7	85.7	8.4	14.3
Overall Accuracy: 94.3%					
Kappa Coefficient: 0.92					

(85.7%, Table III). This might be caused by the disappearance of the absorption features in gasoline soil samples when the gravimetric gasoline content was lower than 8%.

The Gini index derived from the RF classification revealed the top spectral bands for the detection of dry, water, diesel, and gasoline soils. The important spectra were identified at 1040–1103 nm followed by 1610–1694, 2258–2280, 2485–2500, 1200–1250, and 2150 nm (see Fig. 7). The dry soils showed significantly higher albedo, making it distinguishable from the other types. The reflectance spectra of soil samples demonstrate noticeable differences at spectral bands 1040–1103 nm, where the reflectance is the highest for dry soil, followed by water, gasoline, and diesel (see Fig. 7). At 1610–1694 nm, weak absorptions were observed for all types of soil samples except that diesel soils showed significantly lower reflectance than water and gasoline soils. Moreover, the reflectance patterns abruptly changed at ∼1700 nm where water soils showed significantly higher reflectance than gasoline and diesel soils because PHCs have strong C–H bend absorptions. On the other hand, the C–H stretch and C–H bend absorption features at 2250–2300 and >2450 nm caused low reflectance for gasoline and diesel soils distinguishable from water and dry soils (see Fig. 7). Relatively smaller importance was found at 1200–1250 and 2150 nm, where the former separates reflectance difference for all groups, and the latter separates diesel from others (see Fig. 7).

Previous studies on crude oil and PHCs polluted soils have shown C–H bend absorptions at 1645, 1712, 1752, 1758,

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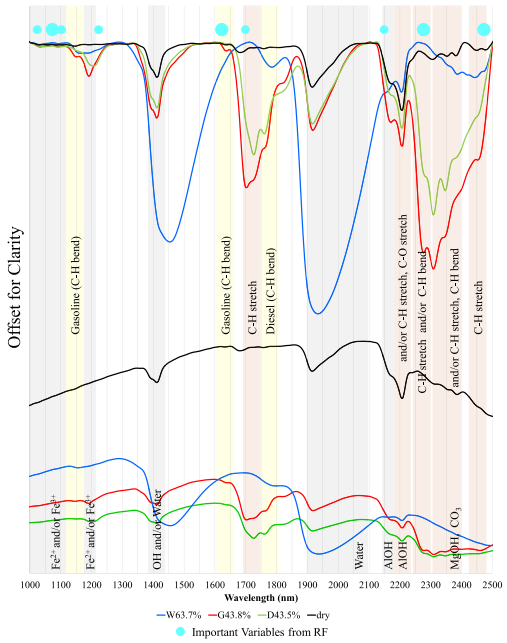


Fig. 7. Top 30 important spectral variables for soil classification derived from RF overlaid with reflectance and hull quotient spectra of dry, water, diesel, or gasoline soils from 1000 to 2500 nm.

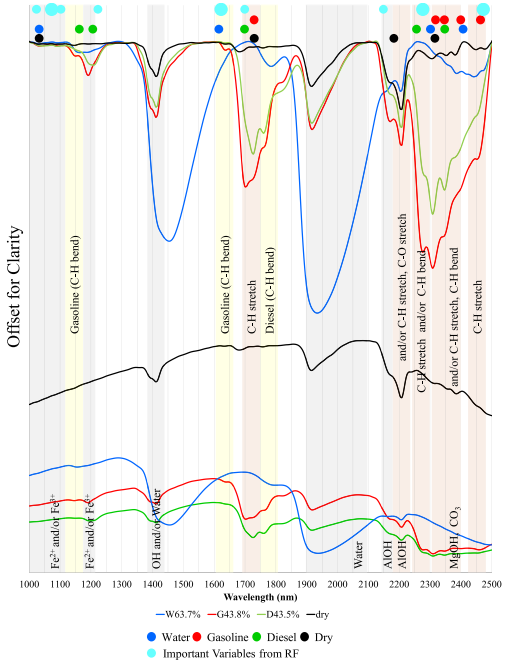


Fig. 8. Important spectral variables for detecting each target derived from the binary logistic regression overlaid with reflectance spectra of dry, water, diesel, or gasoline soils from 1000 to 2500 nm.

2207, 2220, 2240, and 2460 nm [6], [12], [64]. However, they focused on detecting only one type of pollution, either by crude oil or diesel. In the real-world scenarios, we need to consider other soil conditions such as dry clean soil, wet soil, or gasoline-polluted soil. We also found that the integration of spectral bands associated with PHCs, and spectral bands associated with mineral components would increase the detection applicability for the real-world cases.

2) Derivation of Detection Models by Logistic Regression Models: Although the RF showed excellent performance in the detection of the four classes of soils, the RF requires training data for real-world application. Therefore, we developed detection models applicable to real-world cases with binary logistic regressions.

Table IV summarizes the best logistic regression model for each soil contamination condition. The best model for dry soil classification uses four spectral bands: two bands are associated with C–H stretch and C–H bend absorptions, and the other two bands are associated with clay minerals and ferric ions (see Fig. 8 and Table IV). The water soil detection model uses four spectral bands, with three bands from C–H bands and one from ferric ion bands. The diesel detection model uses two ferric ion bands and three C–H bands, and the gasoline detection model uses six C–H bands (see Fig. 8 and Table IV).

Each logistic regression model works effectively to detect the target soil samples. Therefore, each class picked different spectral bands as its best detection model. These bands are recognized as indicator spectral bands for each soil class. For example, in Fig. 8, the indicator spectral bands for diesel soil are the five bands marked by green dots.

The indicator spectral bands to detect dry soils were composed of spectral bands related to mineral components

TABLE IV

SELECTED WAVELENGTHS FOR FOUR TYPES OF SOILS MODELS DERIVED FROM THE BINARY LOGISTIC REGRESSION AND WALD TEST RESULTS

Wavelength(nm)	β [1]	S.E. [2]	Wald [3]	df [4]	Sig. [5]	Exp (β) [6]
Final selected spectral variables for dry soil classification						
1142	-0.009	0	371.329	1	0	0.991
1738	0.009	0.001	296.705	1	0	1.009
2186	-0.025	0.001	889.594	1	0	0.976
2319	0.025	0.001	822.541	1	0	1.025
Constant	-4.211	0.544	59.831	1	0	0.015
Final selected spectral variables for water soil classification						
1142	0.008	0.001	177.353	1	0	1.008
1610	-0.006	0	176.019	1	0	0.994
2308	0.022	0.001	233.098	1	0	1.022
2419	-0.24	0.001	264.197	1	0	0.976
Constant	-5.971	0.848	49.528	1	0	0.003
Final selected spectral variables for diesel soil classification						
1170	0.123	0.008	219.880	1	0	1.131
1215	-0.150	0.010	223.787	1	0	0.861
1688	0.040	0.003	196.899	1	0	1.041
2264	0.042	0.003	193.167	1	0	1.042
2347	-0.081	0.006	202.379	1	0	0.923
Constant	-2.379	0.289	67.614	1	0	0.093
Final selected spectral variables for gasoline soil classification						
1727	0.007	0	609.786	1	0	1.007
2313	-0.020	0.001	252.290	1	0	0.980
2330	-0.64	0.002	1656.381	1	0	0.938
2347	0.044	0.001	1079.882	1	0	1.045
2407	0.059	0.001	1939.248	1	0	1.060
2463	-0.027	0.001	1258.417	1	0	0.973
Constant	-7.057	0.248	812.037	1	0	0.001

β [1] = logistic coefficient; S.E. [2] = standard error of estimate; Wald [3] = Wald chi-square values; df [4] = degree of freedom; Sig. [5] = p-value; Exp(β) [6] = exponentiated coefficient.

associated with clay minerals and ferric ions, which are the most found mineral components in soils [70]. The clay mineral signals are always modified by liquid components such as

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TABLE V
MODEL SUMMARY OF THE BINARY LOGISTIC REGRESSION MODEL IN EACH TARGET

Parameters	Pseudo-R^2		Hosmer and Lemeshow Test		
	Cox and Snell R^2	Nagelkerke R^2	χ^2	df	p-value
Dry	0.652	0.965	11.734	8	0.163
Water	0.634	0.977	6.368	8	0.606
Diesel	0.705	0.999	2.088	8	0.978
Gasoline	0.634	0.960	8.415	8	0.394

TABLE VI
COMPARATIVE ANALYSIS FOR DETECTION MODELS WITH AND WITHOUT MINERAL BANDS

Class	Accuracy (%)		Kappa coefficient	
	With mineral bands	Without mineral bands	With mineral bands	Without mineral bands
Dry	98.99	79.42	0.97	0.26
Water	94.70	72.59	0.92	0.44
Diesel	98.60	80.87	0.96	0.46
Gasoline	72.10	57.75	0.71	0.26

water, diesel, and gasoline added to dry soils. On the other hand, the water detection model uses C–H and ferric ion bands. The gasoline and diesel soil samples show distinctive absorptions at C–H bands, whereas water soil has less effect in the spectral region, showing relatively higher reflectance compared to the PHCs. The diesel detection model uses C–H bands along with the ferric ion band. The C–H bands can separate dry and water soils, and the ferric ion band can separate diesel soils from gasoline soils. Notably, the C–H bending bands at the 2200–2450 and 1727 nm regions can effectively separate gasoline soils from other soils.

All models passed the pseudo-R^2 goodness of fit test with values larger than 0.2 (see Table V) [71]. The results of the Hosmer and Lemeshow test showed that the p-values of all models were greater than 0.05, which indicates that the models are acceptable (see Table V) [72].

To test the detection efficiency of the integration of PHCs and mineral absorptions to PHCs-only models, we have derived additional models using PHCs bands. The accuracy and kappa coefficient of the models using PHCs bands range from 57.8% to 80.9% and 0.26 to 0.46, respectively. They are about 14.4% to 22.1% lower than the models using both PHCs and mineral bands (see Table VI).

3) Validation of the Models and Imaging Soil Types: The binary logistic regression models developed from the training data were applied to the validation data (see Fig. 9). All models showed excellent accuracy ranging from 72.10% to 98.99% with an overall accuracy of 91.82% and kappa coefficient of 0.89. The overall accuracy of the dry soil detection model was 98.99%. Although all dry soil samples were correctly detected, 14.46% of commission errors were found in the areas of water or gasoline classes [see Fig. 9(b) and Table VII]. The commission error was caused by water samples with gravimetric content of less than 0.6% and by gasoline samples with gravimetric content of less than 3.1%. This observation

TABLE VII
CONFUSION MATRIX TABLE OF THE VALIDATION DATA FOR THE CLASSIFICATION RESULTS FROM THE BINARY LOGISTIC REGRESSION MODELS

Class	Validation Data		
	Accuracy (%)	Commission Error (%)	Omission Error (%)
Dry	98.99	14.46	1.01
Water	94.70	0.09	5.30
Diesel	98.60	0.09	1.40
Gasoline	72.10	1.20	27.90
Overall Accuracy: 91.82%			
Kappa Coefficient: 0.89			

was discussed in the spectral analysis on the disappearance of spectral features with a decrease in gasoline content. Moreover, a water content of less than 0.6% can be considered a dry condition resulting in similar spectral features as dry soils. The accuracy of water and diesel soils showed excellent detection results with a small commission and omission errors [see Fig. 9(c) and (d) and Table VII]. The accuracy of gasoline soils detection was relatively lower than the other classes with an overall accuracy of 72.10%. The relatively high omission error (27.9%) of the gasoline detection model was caused by the fast evaporation rate of gasoline.

These results confirm that the detection models developed in this study can be effectively used for the detection of PHC-polluted soils from clean soils including wet and dry soils with high accuracy. Moreover, pixel-based classification on hyperspectral images using these models demonstrates the feasibility of using ground-based or unmanned aerial vehicle (UAV)-based platforms for bundle detection of PHCs pollution in soils. The validated detection models for imaging dry soil, water soil, diesel soil, and gasoline soil are shown at the top of page 11, where $P_{\text{dry soil/ water soil/diesel soil/ gasoline soil}}$ is the probability of each class, and each wavelength stands for the reflectance value at each wavelength.

D. Field Applicability Test

The detection models derived from the logistic regression models were applied to the hyperspectral images for the field applicability test. The models detected the target soils with an overall accuracy of 83.36% and a kappa coefficient of 0.79 (see Fig. 10 and Table VIII). Among the four classes, dry soil detection showed the lowest producer's accuracy of 58.81%. The misclassified pixels were mostly classified as an unclassified class. We think that this relatively higher error was caused by soil conditions. The detection models were derived with preprocessed soils after completely drying and sieving to remove moisture and granule effect. However, the field test used the soils without preprocessing to simulate real-world field conditions. Although the dry soils showed relatively lower accuracy, the model did not misclassify dry soils to the other three classes. It infers that the model is still applicable in the field condition to detect dry soils. It may be

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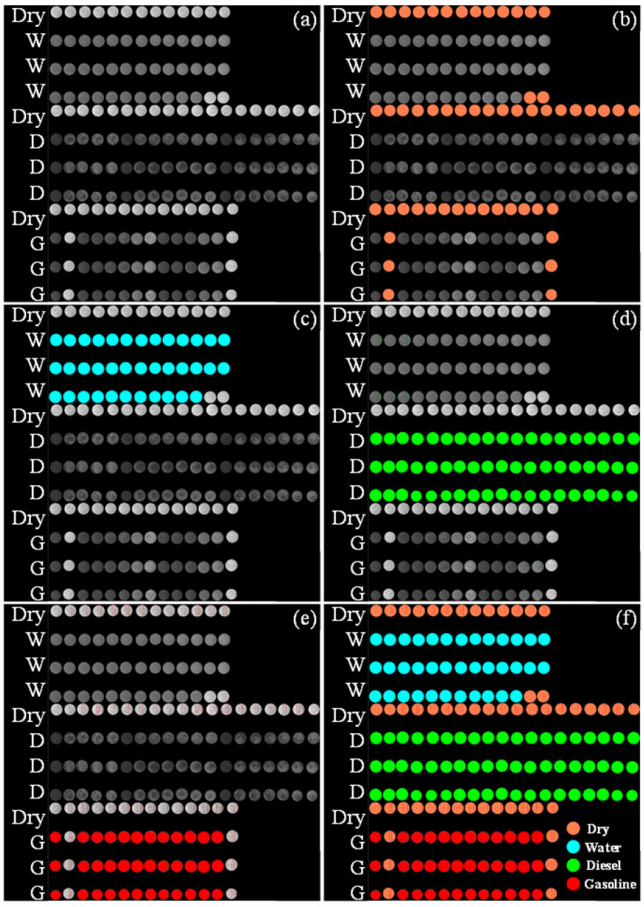


Fig. 9. Classification images of the binary logistic regression models applied to the validation image. (a) Validation image loaded as a grayscale image using the 1002 nm, (b) dry, (c) water, (d) diesel, (e) gasoline, and (f) combination of (b)–(e), where the sample label of dry shows the dry soils, W is the water soils, D is the diesel soils, and G is the gasoline soils.

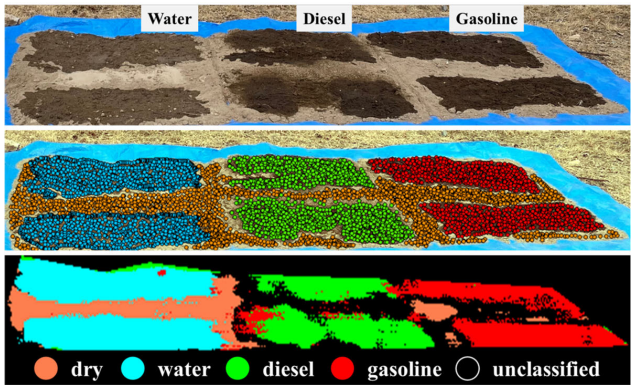


Fig. 10. Field test setting, sample pixel distribution, and classification image of the binary logistic regression models applied to the field test image; coral for dry soils, cyan for water soils, green for diesel soils, and red for gasoline soils.

necessary to drive another model for dry soils with various types of soils without preprocessing.

Different from dry soils, the other three classes were detected with very high accuracy ranging from 90.38% for diesel soils to 99.67% for water soils (see Table VIII). It infers that the detection models for water, diesel, and gasoline soils were less affected by soil properties such as larger particle

TABLE VIII
CONFUSION MATRIX TABLE OF THE FIELD TEST DATA FOR THE CLASSIFICATION RESULTS FROM THE BINARY LOGISTIC REGRESSION MODELS

Class	Validation Data		
	Accuracy (%)	Commission Error (%)	Omission Error (%)
Dry	58.81	0.04	41.19
Water	99.67	11.13	0.33
Diesel	90.38	3.10	9.62
Gasoline	91.18	6.90	8.82
Overall Accuracy: 83.36%			
Kappa Coefficient: 0.79			

sizes. The classification accuracy for the field test showed as excellent as the laboratory experiment for water and diesel soils. Moreover, the gasoline soils were detected with higher accuracy, which was mainly associated with the evaporation rate, where the field test was carried out within a 2-h period after the complete saturation.

This study introduced the models for the detection of diesel and gasoline soils along with dry and water soils, which are the most commonly found PHCs pollutant in everyday life over the world. The detection models derived from laboratory experiments detected all classes with very high accuracy, and the gasoline detection model defined the working range of gravimetric content. The models were also tested in the field condition for feasibility analysis, and the models' detection performance was proved. Given the fact that the soil samples used in this study are one of the representative types in a natural environment with common mineral composition, the models can be applicable in real-world cases. Although our field test used a close-range platform, we think the models can be applied for UAV and aerial platforms because the water vapor absorption and atmospheric scatter bands have been excluded from the model construction. It is expected that our models can bring advancement to remote sensing applications for petroleum-associated pollutions.

IV. CONCLUSION

The article presents the procedure for identifying indicator spectral bands and associated logistic models for detecting diesel or gasoline-polluted soils from wet and dry soils using an SWIR hyperspectral scanner. The mineralogical analysis revealed that soil samples used in the experiments can be considered representative soils originating from the weathering process of temperate climate. Moreover, the soil can be classified as sandy loam soil based on the USDA soil classification system, which is commonly found in the natural environment.

Spectral characteristics of soil samples were manifested by secondary minerals showing absorption features of AlOH, OH, MgOH, CO_3, and ferric ions. The water soil sample spectra showed a systematic increase of water absorption near 1400 and 1900 nm. The overall spectral reflectance of diesel soils gradually decreased in the whole spectral range with an increase in diesel content. The spectral signatures of diesel

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$$P_{\text{dry soil}} = \frac{e^{(-0.009\times 1142\ \text{nm}+0.009\times 1738\ \text{nm}-0.025\times 2186\text{nm}+0.025\times 2319\ \text{nm}-4.211)}}{1+e^{(-0.009\times 1142\ \text{nm}+0.009\times 1738\ \text{nm}-0.025\times 2186\ \text{nm}+0.025\times 2319\ \text{nm}-4.211)}}$$

$$P_{\text{watersoil}} = \frac{e^{(0.008\times 1142\ \text{nm}-0.006\times 1610\ \text{nm}+0.022\times 2308\ \text{nm}-0.024\times 2419\ \text{nm}-5.971)}}{1+e^{(0.008\times 1142\ \text{nm}-0.006\times 1610\ \text{nm}+0.022\times 2308\ \text{nm}-0.024\times 2419\ \text{nm}-5.971)}}$$

$$P_{\text{diesel soil}} = \frac{e^{(0.123\times 1170\ \text{nm}-0.150\times 1215\ \text{nm}+0.040\times 1688\ \text{nm}+0.042\times 2264\ \text{nm}-0.081\times 2347\ \text{nm}-2.379)}}{1+e^{(0.123\times 1170\ \text{nm}-0.150\times 1215\ \text{nm}+0.040\times 1688\ \text{nm}+0.042\times 2264\ \text{nm}-0.081\times 2347\ \text{nm}-2.379)}}$$

$$P_{\text{gasoline soil}} = \frac{e^{(0.007\times 1727\ \text{nm}-0.020\times 2313\ \text{nm}-0.064\times 2330\ \text{nm}+0.044\times 2347\ \text{nm}+0.059\times 2407\ \text{nm}-0.027\times 2463\ \text{nm}-7.057)}}{1+e^{(0.007\times 1727\ \text{nm}-0.020\times 2313\ \text{nm}-0.064\times 2330\ \text{nm}+0.044\times 2347\ \text{nm}+0.059\times 2407\ \text{nm}-0.027\times 2463\ \text{nm}-7.057)}}$$

soils showed integrative spectral characteristics manifested by both mineral absorptions and PHCs. The PHCs absorption features manifested by C–H stretching or bending modes of terminal CH_3 and saturated CH_2 groups systematically increased with an increase in diesel content. The overall spectral patterns of gasoline soils were like diesel soils: PHCs-associated spectral features increase with gasoline content. However, the gasoline soils showed relatively weaker doublet absorptions near 1750 and 2350 nm and stronger doublet absorptions near 2270 and 2450 nm. The differences can be explained by the gasoline production procedure, where gasoline contains a high concentration of aromatic hydrocarbons associated with aromatic C–H bend. Notably, the spectral features associated with PHCs in gasoline soils disappeared below 8%, which could be a detection limit using spectral variables.

The RF model showed that the spectral signals acquired from the hyperspectral camera can be effectively used for the detection of water, diesel, and gasoline soils, showing classification accuracy and kappa coefficient of 94.3% and 0.92%, respectively. However, the producer's accuracy for gasoline soils was relatively lower (85.7%) because of the detection limit. The spectral variables, which played important roles in the classification, were 1040–1103 nm of iron oxide followed by 1610–1694 nm of C–H bending, 2258–2280 nm of C–H bending, and 2485–2500 nm of BTEX C–H. It indicates that iron oxide and PHC bands are important for the detection of diesel and gasoline contamination.

The dry soil logistic model was built with two ferric ion/clay mineral indicator bands and two C–H indicator bands. The water soil model uses four indicator bands: one ferric ion band and three C–H bands. The diesel model uses five indicator bands: two ferric ion bands and three C–H bands. The gasoline model uses six C–H indicator bands. All logistic models with indicator spectral bands showed an excellent detection accuracy from 72.1% to 98.99% with an overall accuracy of 91.82% and kappa coefficient of 0.89. The field applicability test of the detection models for unprocessed soils showed an accuracy of 83.36% and kappa coefficient of 0.79.

An important finding from our analyses is that the integration of spectral bands associated with PHCs and mineral components would increase the detection accuracy for diesel and gasoline pollution in soil. Specifically, we excluded water absorption bands from the detection models to avoid water vapor absorption in general remote sensing applications. Overall, the research is a pilot study of deriving spectral indicators for diesel and gasoline contamination detection in field conditions. Although our field test used a close-range platform, it can help in developing UAV or airborne applications for large-scale soil pollution detection, as foreseen by Asadzadeh et al. [73].

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